Control Theory

Linear Algebra Preparement

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Objectives:

1. Master how to calculate the Eigenvalues of Matrix $A \in \mathcal{R}^{n\times n}$, especially by Software
2. Master how to calculate the Inverse, Decomposition(Eigenvalue and Jordan Forms) of Matrix $A \in \mathcal{R}^{n\times n}$, especially by Software
3. Master how to determine the positiveness, negativeness of Matrix $A \in \mathcal{R}^{n\times n}$, especially by Software
4. Master the basic math according of Matrix Exponential calculation e^{At},especially by Software

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Contents

1. Linear Algebra: Vector and Matrix
2. Matrix Decomposition
3. Positive Definite Matrix
4. Matrix Exponential e^{At}



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For a group of equations, find the solutions of u, v, w

$$\begin{aligned} a_{11}u + a_{12}v + a_{13}w &= b_1 \\ a_{21}u + a_{22}v + a_{23}w &= b_2 \\ a_{31}u + a_{32}v + a_{33}w &= b_3 \end{aligned} \tag{1}$$

where $a_{i,j}$ are known constant scalar values.
(1) can be further expressed as

$$Ax = b \tag{2}$$

where vectors x, b and matrix A are

$$x = \begin{bmatrix} u \\ v \\ w \end{bmatrix}, b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}, A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$



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Theorem

When the Determinant function $|A|$ is non-zero , the solution for x in (2) is unique.

Definition (Determinant)

$$
\begin{aligned}
|A| &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\
&= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}
\end{aligned}
$$

where

$$
\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc
$$



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When $|A| \neq 0$, it has that

$$Rank(A) = n \tag{3}$$

n is the dimension of matrix A.

The solutions of λ for

$$|\lambda I - A| = 0 \tag{4}$$

$\lambda_1, \lambda_2, \cdots$, and λ_n are non-zero.

Definition (Eigenvalues)

$\lambda_1, \lambda_2, \cdots$, and λ_n are the eigenvalues of matrix A.

Definition (Inverse)

If the two matrices A and $M \in \mathcal{R}^{n \times n}$

$$AM = I \tag{5}$$

one defines that A (or M) is the inverse matrix of M (or A), which can be expressed as $A^{-1} = M$ (or $M^{-1} = A$).

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When solving a system of linear equations in (2),

$$Ax = b$$

it has that, for any given matrix $L \in \mathcal{R}^{m\times n}$

$$LAx = Lb \tag{6}$$

if L is invertible, $L \in \mathcal{R}^{n\times n}$ and rank of $(LA) = n$ the solutions of x in (6) and (2) are same.

Define $x = R\eta$, $R \in \mathcal{R}^{n\times n}$

$$LAR\eta = Lb \tag{7}$$

the solution of $x(t)$ can be obtain via firstly finding the solution of η.



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Eigendecomposition

$$A = VDV^{-1} \tag{8}$$

where D is a diagonal matrix formed from the eigenvalues of A, $\lambda_1, \lambda_2, ..., \lambda_n$, and the columns of V are the corresponding eigenvectors of A.

Jordan-decomposition

$$A = SJS^{-1} \tag{9}$$

where J is a Jordan matrix from the eigenvalues of A, $\lambda_1, \lambda_1, \cdots, \lambda_2, \lambda_2, \cdots, ..., \lambda_n$.

Singular Value Decomposition (SVD)

For a given matrix $A \in \mathcal{R}^{n\times p}$, there are matrices $U \in \mathcal{R}^{n\times n}$, $S \in \mathcal{R}^{n\times p}$ and $V \in \mathcal{R}^{p\times p}$, that

$$A = USV^T \tag{10}$$

where $U^TU = I_{n\times n}$, $V^TV = I_{p\times p}$.

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For any scalar values a and b, one obtains

$$a^2 + b^2 - 2ab \geq 0 \tag{11}$$

or $a \neq b$, and for given scalars α, β and γ, if

$$\begin{aligned} & a^2\alpha + b^2\beta - 2\gamma ab \\ = & \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} \alpha & -\gamma \\ -\gamma & \beta \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \\ > & 0 \end{aligned} \tag{12}$$

one can say $\begin{bmatrix} \alpha & -\gamma \\ -\gamma & \beta \end{bmatrix}$ is positive.

Similarly,

Definition

M is **positive definite** $\Longleftrightarrow x^T Mx > 0$ for any $x \in \mathcal{R}^n \neq 0$.
M is **positive semi-definite** $\Longleftrightarrow x^T Mx \geq 0$ for any $x \in \mathcal{R}^n \neq 0$.

Lemma

Let M be an $n \times n$ Hermitian matrix. Its eigenvalues are real, such that

- *M is positive (or negative) definite if and only if all of its eigenvalues are positive (or negative) .*
- *M is positive (or negative) semi-definite if and only if all of its eigenvalues are non-negative (or non-positive) .*
- *M is indefinite if and only if it has both positive and negative eigenvalues.*

Lemma

Sum of two matrices: M and N are with same dimensions

- *If M and N are positive-definite , then the sum $M + N$ is also positive-definite .*
- *If M and N are positive-semi-definite , then the sum $M + N$ is also positive-semi-definite .*
- *If M is positive-definite and N is positive-semi-definite, then the sum $M + N$ is also positive-definite .*

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Exponential function, Series expansion

$$e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!} \tag{13}$$

and

$$e^{at} = \sum_{i=0}^{\infty} \frac{a^i t^i}{i!} \tag{14}$$

For the Laplace transform

$$\mathcal{L}\left(e^{-at}\right) = \frac{1}{s+a} = (s+a)^{-1} \tag{15}$$



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Definition

Matrix Exponential: the matrix exponential of $A \in \mathcal{R}^{n\times n}$, converges absolutely for any time t.

$$\begin{aligned} e^{At} &= \sum_{i=0}^{\infty} \frac{A^i t^i}{i!} \\ &= e^{P\lambda P^{-1}t} = \sum_{i=0}^{\infty} \frac{(P\lambda P^{-1})^i t^i}{i!} \\ &= P\sum_{i=0}^{\infty} \frac{\lambda^i t^i}{i!} P^{-1} = Pe^{\lambda t}P^{-1} \\ &= \mathcal{L}^{-1}\left[(sI-A)^{-1}\right] \end{aligned} \tag{16}$$

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- $\Phi(0) = e^{A0} = I$
- $\Phi(t) = e^{At} = (e^{-At})^{-1} = (\Phi(-t))^{-1}$ or $\Phi(-t) = \Phi^{-1}(t)$
- $\Phi(t_1+t_2) = e^{A(t_1+t_2)} = e^{At_1}e^{At_2} = \Phi(t_1)\Phi(t_2) = \Phi(t_2)\Phi(t_1)$
- $[\Phi(t)]^n = \Phi(nt)$
- $\Phi(t_2-t_1)\Phi(t_1-t_0) = \Phi(t_2-t_0) = \Phi(t_1-t_0)\Phi(t_2-t_1)$



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谢谢!

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